

Data Structure Lab9 : Positional Linked List and Array List 2024-2025

Topics

1. Create Position Interface
2. Create Positional List interface
3. Create Positional Linked List Using Linked List structure (Nodes)
4. Implement Basic Methods of Positional Linked List
 - addBefore(Position<E> p ,E e)
 - addAfter(Position<E> p ,E e)
 - remove(Position<E> p)
5. Implement Iterator and Iterable pattern design in Positional Linked Lists

Homework

1. Implement the ArrayList Data structure as it is described in chapter 7.

```
import java.util.Arrays;
```

```
public class ArrayList<E> {  
    private E[] array;  
    private int size;  
    private static final int DEFAULT_CAPACITY = 10;  
  
    public ArrayList() {  
        array = (E[]) new Object[DEFAULT_CAPACITY];  
        size = 0;  
    }  
  
    public void add(E element) {  
        ensureCapacity();  
        array[size++] = element;  
    }  
}
```

Data Structure Lab9 : Positional Linked List and Array List 2024-2025

```
    }

    public E get(int index) {
        if (index < 0 || index >= size) throw new
IndexOutOfBoundsException();
        return array[index];
    }

    public int size() {
        return size;
    }

    private void ensureCapacity() {
        if (size == array.length) {
            array = Arrays.copyOf(array, array.length * 2);
        }
    }

    public boolean isEmpty() {
        return size == 0;
    }
}
```

2. Implement the iterator idea in your ArrayList.

```
import java.util.Iterator;

public class ArrayList<E> implements Iterable<E> {
    // Existing ArrayList code...

    @Override
```

Data Structure Lab9 : Positional Linked List and Array List 2024-2025

```
public Iterator<E> iterator() {  
    return new ArrayIterator();  
}  
  
private class ArrayIterator implements Iterator<E> {  
    private int current = 0;  
  
    @Override  
    public boolean hasNext() {  
        return current < size;  
    }  
  
    @Override  
    public E next() {  
        if (!hasNext()) throw new NoSuchElementException();  
        return array[current++];  
    }  
}
```